Report for Question 4 in Deep Learning Coursework

Basic setting of experiment

In this study, three models were evaluated: a deterministic baseline model and two Bayesian neural network approaches, namely MC Dropout and SWAG. Both the baseline and MC Dropout models were trained for 200 epochs using stochastic gradient descent (SGD) with a learning rate of 0.001 and a momentum of 0.9. For the MC Dropout model, the dropout rate was set to 0.2. To construct the SWAG model, a second training phase was conducted by fine-tuning the best-performing deterministic model for an additional 50 epochs. During this phase, the SGD hyperparameters were adjusted to a higher learning rate of 0.01 and a reduced momentum of 0.1. The larger learning rate encourages exploration of the local weight space, which helps SWAG approximate the posterior distribution of model parameters.

Experiment results

We first evaluate the models using a standard single forward pass on the test set. For the Bayesian models (MC Dropout and SWAG), this means predicting without marginalizing over the parameter uncertainties.

Model (Single Pass)	**Test Loss**	**Test Accuracy**
Standard CNN (Deterministic)	0.7106	**81.96%**
MC Dropout (No Ensemble)	0.8024	77.23%
SWAG (Single Sample)	**0.5964**	81.37%

Table 1: Comparison of Test Loss and Test Accuracy for single-pass evaluations (without Bayesian ensembling) on the test set.

As shown in Table 1, the deterministic baseline achieves the highest single-pass accuracy. The MC Dropout model performs poorly in this setting, as the active dropout layer randomly masks crucial features during the single prediction

Expected calibration error (ECE)

We summarize several evaluation metrics for the deterministic baseline and the two Bayesian models.

Model	**Test Loss**	**Test Accuracy**	**ECE**
Standard CNN (Deterministic)	0.7106	81.96%	0.0539
Predictive Distribution ($S = 50$, MC Dropout)	0.5818	79.84%	0.0180
Predictive Distribution ($S = 50$, SWAG)	**0.4865**	**83.79%**	**0.0141**

Table 2: Comparison of Test Loss, Test Accuracy, and Expected Calibration Error (ECE) across the baseline deterministic CNN, MC Dropout, and SWAG models.

As ECE is given by

$$\sum_{n=1}^{m} \frac{B_m}{N} |\text{acc}(B_m) - \text{conf}(B_m)|. \tag{1}$$

Ideally, the Expected Calibration Error (ECE) should be as close to zero as possible. The ECE values for both MC Dropout and SWAG are remarkably low, indicating excellent model

calibration, whereas the baseline deterministic model exhibits a significantly higher ECE. Between the two Bayesian approaches, SWAG achieves a slightly lower ECE than MC Dropout, demonstrating the best overall calibration among the evaluated models.

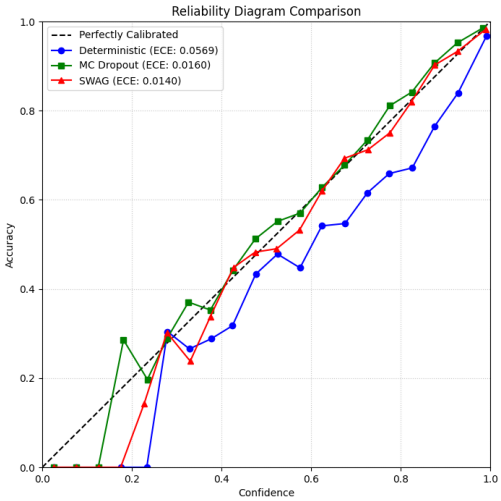


Figure 1: Accuracy vs Confidence. The experiments are done by choosing $S = 30$.

Comments and analysis: A perfectly calibrated model is represented by the diagonal dashed line, indicating that the model's predictive confidence aligns exactly with its empirical accuracy. As shown in the figure, the benchmark deterministic CNN exhibits overconfidence, with its curve falling significantly below the diagonal. This implies that the model frequently outputs unjustifiably high probabilities even for incorrect predictions, which is highly undesirable in real-world applications. Conversely, the Bayesian approaches (MC Dropout and SWAG) successfully mitigate this issue, though MC Dropout shows slightly underconfidence. By incorporating model uncertainty, they produce much more reliable probability estimates. Their calibration curves closely track the ideal dashed line, demonstrating that these BNNs are exceptionally well-calibrated and appropriately lower their confidence when facing uncertain samples.

Conclusion

Based on the experimental results, the Bayesian approaches demonstrate significantly better calibration than the deterministic baseline, as their predictive confidence more closely aligns with empirical accuracy. In contrast, the deterministic CNN tends to produce overconfident predictions. In terms of predictive performance, the MC Dropout model shows a reduction in accuracy compared to the baseline, likely due to the stochastic masking introduced during inference. However, the SWAG model overcomes this limitation and achieves the best overall performance in our experiments, combining high predictive accuracy with strong calibration.